

Validation Glossary

Plain-language reference for the case classification validation project. Enough to follow a conversation or read the schema without digging through the design documents.

A model reads a support case and says which **skill** it belongs to, what **object failed** and what the **defect** was. Before that goes into the product, we have to know how often it is right.

A second, independent model reads the same cases and answers separately. Where the two agree, we accept the answer automatically. Where they disagree, a subject matter expert decides. A small sample of the agreed cases is checked by hand anyway, to measure how often agreement was wrong. Where the experts find the fault is in our own skill wording rather than the model, the wording is rewritten, both models are rerun, and the cases that changed are checked again.

People and systems WHO DOES WHAT

Data engineer	Prepares the fixed case sample, loads the reference data, and runs the job that decides which cases need a person. Owns the database.
Data science team	Runs the models. Receives case numbers and skill definitions, returns predictions.
Primary model run_role = primary	The LLM being validated. The one that would go into the product.
Second opinion run_role = second_opinion	An independent LLM, ideally a different model or vendor, that answers the same cases blind . Its job is to find cases worth a human's time, not to be the final word.
SME	Subject matter expert. An experienced engineer who decides which answer is right when the two models disagree.
Supervisor	An SME who can see across all products, reassign cases and release stuck locks.
Review form	The screen an SME works in. Shows the case beside both model answers and records the verdict.
Triage job	Scheduled job that compares the two model answers and decides, case by case, what is accepted automatically and what is queued for an SME.

What the model produces THE THREE ANSWERS PER CASE

Skill skill_code	The category the issue belongs to, from a fixed list. This is what routing and reporting are built on.
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Failing object object_failed	The thing that broke, in the model's own words: "Controller B", "uplink port 1/3". Free text, so comparing two answers needs an equivalence check rather than an exact match.
Defect defect	What went wrong with that object: "boot loop after upgrade". Also free text.
Rank 1	The model's chosen skill. It may return alternates at rank 2 and 3, which is why the form asks whether the right skill was ranked first.
Explanation	The model's reasoning, shown to the SME. Often the quickest way to see whether a wrong answer is a model problem or an unclear definition.
Confidence	The model's own score, 0 to 1. Used as a triage signal, not as evidence of correctness.

The skill taxonomy WHAT WE ARE ASKING THE MODEL TO APPLY

Skill definition	The detailed wording for one skill, with what it includes and excludes. Sent to the data science team alongside the case numbers.
Taxonomy version TAX-1.0, TAX-1.1	A dated release of all definitions. Every model run records which version it used, so a case is always judged against the wording in force at the time.
Definition change	A request from an SME to reword a skill, raised from the case that prompted it. Approved changes go into the next taxonomy version.
Definition gap	The model's answer was defensible because our wording didn't separate two skills clearly. A taxonomy problem, not a model problem.

Running and checking HOW A CYCLE WORKS

Cycle	One pass: run both models, triage, SME review, report, agree definition changes. Two or three cycles are planned.
Run run_id	One execution of one model over the test set. A cycle has two: primary and second opinion.
Blind	Answering without seeing anyone else's answer first. The second model is always blind; SMEs are blind on calibration and audit cases. Seeing another answer first makes people and models agree with it more than they should.
Triage	Comparing the two answers field by field and deciding the outcome: accept automatically, or send to an SME.
Auto-accepted	Both models agreed on all three fields and both were confident. No person looks at it. Around 80% of cases.

Confidence threshold	The setting that decides how much gets auto-accepted. Raising it sends more to people and lowers the error rate in the accepted pile. The 80% is a consequence of this setting, not a fixed target.
Calibration set is_golden_set	About 300 cases labelled blind by SMEs before we trust the automation, to measure how accurate the second model actually is. Previously called the golden set.
Audit sample	A random few per cent of auto-accepted cases, reviewed by an SME anyway. The only way to know how often the two models were wrong together.
Adjudication	An SME deciding, where the models disagree, which answer is right — or that neither is.
Rerun	Running both models again on updated definitions. Only cases whose answers changed, or that were wrong last time, are reviewed again.

The review queue HOW SME WORK IS ORGANISED

Queue row review_queue	One case waiting for, or being worked on by, an SME. Carries status, priority and the reason it was queued.
Batch review_batch	All the queued cases for one product in one run. Opened, paused and closed as a unit, which is how products are started one at a time rather than all six at once.
Routing reason	Why this case needs a person: models disagreed, low confidence, audit sample, or calibration. Lets us report which rule actually finds errors.
Entitlement	Which products an SME is allowed to review. Dated, so coverage can change between cycles.
Claim / lock	An SME takes the next case and it locks to them, so two people never get the same one. Abandoned locks are released automatically.
Review slot	Slot 1 is the normal review. Slot 2 is a second SME on the same case, used on a small sample to check that reviewers agree with each other.
Skip vs escalate	Skip puts the case back for someone else. Escalate flags it as genuinely unjudgeable, which is itself a finding.
Superseded	A queue row made obsolete by a rerun. Marked, never deleted.

Verdicts and results WHAT COMES OUT OF IT

Verdict review_verdict	One SME's answers on one case: was the skill right, the object, the defect, and which model they sided with. Never edited — a correction is a new row pointing at the old one.
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Sided with	Primary, second opinion, both or neither. Over a cycle this shows which model is more often right when they disagree.
Error category	Why the answer was wrong: model error (our wording was clear, the model still missed), definition gap (our wording was ambiguous), or data gap (the case notes didn't contain enough to answer). This decides whether the fix is model tuning or rewriting definitions.
Can't tell	A third answer alongside yes and no, for when the case data doesn't support a judgement. Forcing a yes or no here would hide a data problem as a model problem.
Inter-rater agreement	How often two SMEs reviewing the same case reach the same verdict. If it's low, the definitions are unclear — no amount of model tuning will fix that.
Audit error rate	How often the auto-accepted answers turn out to be wrong, measured from the audit sample. The number that tells you whether the automation can be trusted.

Identifiers you'll see NAMING CONVENTIONS

CS-0400001

Case number. The identifier for a case everywhere in the system.

RUN-C2

Model run. The C is the cycle number.

TAX-1.1

Taxonomy version, a dated release of the skill definitions.

B-2-STG9000

Batch: cycle 2, Storage product. One per product per run.

Q2-000148

Queue row: cycle 2, sequence number.

V2-0091

Verdict, one per completed review.

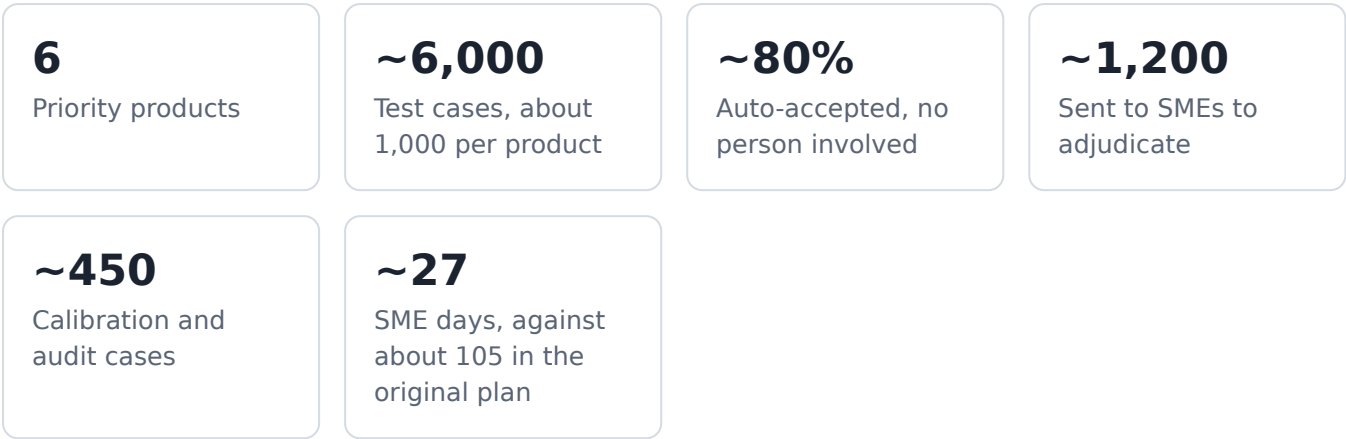
SK-FIRM

Skill code. Permanent, even when the definition is reworded.

DC-014

Definition change request.

The numbers CYCLE 1, ILLUSTRATIVE



The split depends on how often the two models agree and where the confidence threshold is set. The real figures come from the calibration run.

Three rules that explain most of the design IF YOU REMEMBER NOTHING ELSE

Nothing is overwritten	Verdicts and model output are only ever added to. It's the only way to compare one cycle against the next and show whether accuracy improved.
Agreement is not correctness	Two models reading the same ambiguous definition can make the same mistake. That's what the audit sample measures, and why it can't be dropped to save time.
Wrong answers have three causes	The model, our definitions, or the case data. Separating them is what turns an accuracy percentage into a decision about what to fix next.